

Exercise 4: Divergences and Distances Between Probability Measures

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Problem 1 (Computing and Interpreting Kullback–Leibler Divergence):Let P and Q be two discrete probability distributions over the same alphabet $\{A, B, C\}$, defined as

$$P(A) = 0.5, \quad P(B) = 0.4, \quad P(C) = 0.1,$$

$$Q(A) = 0.3, \quad Q(B) = 0.3, \quad Q(C) = 0.4.$$

The Kullback–Leibler divergence from P to Q is defined as

$$D_{\text{KL}}(P\|Q) = \sum_{x \in \{A, B, C\}} P(x) \log \frac{P(x)}{Q(x)}.$$

1. Compute:

$$(a) \ D_{\text{KL}}(P\|Q)$$

$$(b) \ D_{\text{KL}}(Q\|P)$$

Different Weighting: The expectation (average) is calculated weighted by the first distribution (P). If you swap them to calculate $D_{\text{KL}}(Q\|P)$, you are weighting the sum by $Q(x)$ instead of $P(x)$, and the logarithmic term flips to $\ln(Q/P)$.

2. Verify non-negativity. Show that both values are ≥ 0 .**3. Comment on asymmetry.** Explain why $D_{\text{KL}}(P\|Q) \neq D_{\text{KL}}(Q\|P)$ and what this implies about using one distribution to approximate the other.**4. Interpretation.** Based on your numerical results, which distribution is “more surprised” by samples drawn from the other?

$D_{\text{KL}}(Q\|P)$ measures the expected information loss (or surprise) when distribution P is used to model data that actually comes from Q .

Problem 2 (Wasserstein-1 and TV Distance):Let $(\mathcal{X}, d)$ be the discrete metric space

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Given two probability measures μ, ν on $\mathcal{X}$, recall that

$$W_1^{(d)}(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \sum_{x, y \in \mathcal{X}} d(x, y) \pi(x, y),$$

μ 和 ν 的联合概率分布

where $\Pi(\mu, \nu)$ is the set of couplings of μ and ν , and that

$$\text{TV}(\mu, \nu) = \sup_{A \subseteq \mathcal{X}} |\mu(A) - \nu(A)| = \frac{1}{2} \sum_{x \in \mathcal{X}} |\mu(x) - \nu(x)|.$$

1. Show that under the discrete metric d , any coupling π contributes exactly

$$\sum_{x \neq y} \pi(x, y) = 1 - \sum_x \pi(x, x).$$

2. Deduce that

$$W_1^{(d)}(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \left(1 - \sum_x \pi(x, x) \right) = 1 - \sup_{\pi \in \Pi(\mu, \nu)} \sum_x \pi(x, x).$$

3. Finally, prove that

$$1 - \sup_{\pi} \sum_x \pi(x, x) = \frac{1}{2} \sum_x |\mu(x) - \nu(x)| = \text{TV}(\mu, \nu).$$

Conclude that

$$W_1^{(d)}(\mu, \nu) = \text{TV}(\mu, \nu).$$

Problem 3 (Wasserstein distance between two Gaussians):

Let

$$\mu_i = \mathcal{N}(m_i, C_i), \quad i = 1, 2,$$

be non-degenerate Gaussian measures in $\mathbb{R}^n$. Show that

$$W_2(\mu_1, \mu_2)^2 = \|m_1 - m_2\|_2^2 + \text{tr}\left(C_1 + C_2 - 2(C_2^{1/2} C_1 C_2^{1/2})^{1/2}\right).$$